

Modeling the effect of land use and climate change scenarios on future soil loss rate in Kasilian watershed of northern Iran

Mohammad Zare¹ · Majid Mohammady² · Biswajeet Pradhan^{3,4} 

Received: 6 May 2016 / Accepted: 4 April 2017 / Published online: 11 April 2017
© Springer-Verlag Berlin Heidelberg 2017

Abstract Accelerated erosion processes caused by global climate and land use changes in many regions of the world constitute a major restrictive factor in their sustainability. This study proposes a method to estimate soil loss rate under changes in future land use and climate in Kasilian watershed of northern Iran within two periods. The first period is related to current climate and land use (1991–2010), and the second concerns climate and land use scenarios (2011–2030). Downscaling global climate model projections of future climate was applied at the regional scale. A statistical downscaling model was then used to downscale precipitation for three scenarios, i.e., 10% increase in rainfall, 10% decrease in rainfall, and unchanged rainfall. Next, cellular automata–Markov model was used for characterization based on two scenarios of land use future that were designed using suitability maps. The

soil loss mean for the current period was found to be $6.3 \text{ t ha}^{-1} \text{ year}^{-1}$, thereby indicating a low sustainability of soils. The results of simulated soil loss maps indicate a similar pattern in spatial distribution of loss rates compared with those of current periods, but the amount of risk has increased such that simulated erosion mean was 31–58% higher than the current period in all scenarios. Soil loss is thoroughly influenced by climate and land cover patterns in future. In other words, rainfall erosivity has increased by $20 \text{ MJ mm ha}^{-1} \text{ h}^{-1} \text{ year}^{-1}$, based on unchanged rainfall scenario and National Centers for Environmental Prediction data, simulated that cover management factor has increased by 35% compared with the current period. However, simulations indicated that land use changes may potentially induce much larger changes in erosion. The results also showed that soil loss is closely related to land use change and various scenarios of climate change and that revised universal soil loss equation is suitable model to investigate these relationships.

✉ Biswajeet Pradhan
Biswajeet24@gmail.com; biswajeet@lycos.com

Mohammad Zare
Mzare91@ut.ac.ir

Majid Mohammady
majid.mohammady@semnan.ac.ir

Keywords Downscaling · SDSM · CA–Markov · Soil loss · GIS · Kasilian watershed

¹ Faculty of Natural Resources, University of Tehran, Tehran, Iran

² Department of Range and Watershed Management Engineering, College of Natural Resources, Semnan University, Semnan, Iran

³ Department of Civil Engineering, Geospatial Information Science Research Center (GISRC), Faculty of Engineering, University Putra Malaysia, 43400 Serdang, Selangor Darul Ehsan, Malaysia

⁴ Department of Energy and Mineral Resources Engineering, Choongmu-gwan, Sejong University, 209 Neungdong-ro Gwangjin-gu, Seoul 05006, Republic of Korea

Introduction

Soil loss is determined by climatic regime, topography, soil characteristics, land cover, and land use; it can be modified by human activities (Alkharabsheh et al. 2013; Biro et al. 2013). Different studies have indicated that climate and land use factors have the greatest effect among all factors of soil loss (Zhang et al. 2009; Sharma et al. 2011; Quan et al. 2011; Routschek et al. 2014; Litschert et al. 2014; Simonneaux et al. 2015). There are various downscaling

techniques for investigating future climate change and estimation of climate parameters such as precipitation, temperature, and others. In the literature, Nearing (2001), Pruski and Nearing (2002), Plangoen and Babel (2014) applied the delta change method to generate future precipitation. Their results showed that this model has enough accuracy as a downscaling technique. Dibike and Coulibaly (2005) and Hassan et al. (2014) compared LARS-WG and statistical downscaling model (SDSM) downscaling techniques. Their results showed that the SDSM has a better performance when compared to LARS-WG. In another paper, Maeda et al. (2010) showed that climate and land use change have important effect on erosion in the Tanzania and Kenya. Several scenarios were designed to simulate climate and land use in the future, and results explained that about 60 percent of the area will be agriculture by year 2030. Also their results predicted that erodibility will be increased and these factors cause increase of erosion. In the research of Liu et al. (2011), SDSM and bilinear-interpolation (delta) methods were utilized to generate daily series in the yellow river basin. The results showed that SDSM is more accurate for downscaling of climate parameters. Plangoen et al. (2013) investigated effects of climate and land use change on erosion in the part of Thailand, and their results showed that erosion will be increased by year 2040. Stanchi et al. (2013) created a map showing susceptibility to soil erosion in the Aosta Valley (northwest Italian Alps) using chemical and physical properties of soil. Climate, geology, vegetation, and relief were considered as soil-forming factors. Stanchi et al. (2014) applied RUSLE and winter factor to investigate soil erosion in the winter time by the snow cover. Their results showed the relevant role of winter in soil erosion. The inclusion of a winter factor in RUSLE seems promising for the improvement of soil erosion estimates in Alpine environments effected by snow movements. Results of Mondal et al. (2015) showed that erosion will be increased about 18 and 58% by years 2050 and 2080 because of climate and land use change.

Generally, climatic factors are under the effect of global change, and land cover is severely affected by regional change (Wilson et al. 2015); these factors share a complicated relationship. Rise in global temperature leads to an increase in water vapor in atmosphere; such an increase affects regional weather circulation patterns (Mullan et al. 2012). These altered circulation patterns will lead to a changed occurrence of extreme rainstorms as regards duration, frequency, and intensity in the year. Soil loss is mostly the consequence of extreme but short rainfall events and the influence of future rainfall erosivity (Routschek et al. 2014). The potential effects of climate change on soil erosion have been investigated using several approaches (Imeson and Lavee 1998; Neal et al. 2005;

Zhang et al. 2009, 2016); Litschert et al. 2014; Hoomehr et al. 2016). Another effect of climate change is the change in the amount of plant biomass (Klile and Eitziger 2010). The mechanisms in which climate change can affect plant biomass and biomass can affect soil erosion are highly complex because the amount of biomass, in addition to being influenced by climate change, is caused by land use change. The vegetation cover reduction can increase erosion so that vegetation controls soil erosion via its canopy, roots, and litter components (Plangoen et al. 2013). The effect of land use and land cover change on soil loss has been widely studied (Sharma et al. 2011; Quan et al. 2011; Leh et al. 2011; Wijitksum 2012; Li et al. 2014; Khosrokhani and Pradhan 2014; Pradhan et al. 2012; Rizze et al. 2016). Therefore, understanding of current and future status of soil loss in catchment scale is one of the major challenging tasks among natural resource managers. Thus, if future climate and land use can be predicted, soil loss rate in future will be estimated accurately. The revised universal soil loss equation (RUSLE) was developed based on substantial modifications of the USLE and its database to more accurately estimate soil erosion (Renard et al. 1997). Furthermore, RUSLE has been one of the most useful models to estimate soil loss in various studies (Chen et al. 2011; Xu et al. 2012; Ranzi et al. 2012; Prasannakumar et al. 2013; Pan and Wen 2014; Jiang et al. 2015; Karamesouti et al. 2016). To facilitate future soil loss prediction, two significant parameters of rainfall erosivity (R) and land cover management (C) in RUSLE were simulated. Similar to most of the countries, climate change in Iran and especially in the north has affected soil erosion. Also land use change is an important factor to increase erosion in the north of Iran. In recent year, Kasilian watershed has been selected as a watershed index in the north of Iran and many researchers and institutions are studying in this watershed. There is no simultaneous study of climate and land use change in this area and its effects on erosion and soil loss. So the aim of this study is to investigate the effects of these phenomena in the Kasilian watershed. Soil erosion is a severe issue that poses a considerable threat to the ecological environment and human life at the Kasilian watershed in northern Iran. This area is dominated by the Alborz mountain range, in which the ecosystem is highly susceptible to soil erosion because of deforestation, land reclamation, and other inappropriate land use practices. In recent years, a rapid growth in population has occurred in the upper Kasilian watershed. Thus, a large area of grassland and forest has been reclaimed to feed the population, thereby resulting in severe soil erosion and ecosystem damage. An awareness of future soil loss condition under the effect of climate and land use change greatly contributes to Kasilian watershed management.

Study area

The application site for this study is the region surrounding the Caspian Sea (Fig. 1), located at 53°01' to 53°26' E and 35°01' to 36°32' N containing Shirgah City and Zirab City in Mazandaran Province; this site is drained by the Talar River and its tributaries. The site has a spatial extent of almost 342.86 Km² and a mean altitude of 1080 m; the climate is characterized by extreme wet (summer) and dry (winter) seasons with 95% of its annual rainfall occurring between September and June. Mean annual rainfall varies from 600 to 110 mm throughout the catchment with higher rainfall occurring in the plain areas. The average annual temperatures range from 18 °C on the plain area to 10 °C in the mountains. The study area is composed of various land cover types, mainly forest (82.2%), with the remainder being 14% rangeland, 2.4% settlement, and 1.4% agriculture. The soil in this basin is primarily of podzolic, brown forest, and sedimentary types. The Kasilian region is sensitive to environmental change. Soil loss in Kasilian, which has been a critical issue during the past few decades, has been caused by both natural factors and human activity.

Methodology

The proposed method enables assessment of soil loss rate for the current (2000–2010) and future (2011–2030) climate and land use scenarios and can be divided into three steps. (1) Daily rainfall data for future period are generated under three scenarios, namely a 10% increase in rainfall, unchanged rainfall, and a 10% decrease in rainfall. The effects of these scenarios on rainfall erosivity are determined. (2) Future land use change is simulated under the trend of land use demand and multi-criteria evaluation (MCE) scenarios using cellular automata–Markov (CA–Markov). (3) Land use and climate change scenarios are integrated, and new scenarios are created to simulate soil loss rate in future (2011–2030). The process steps are shown in Fig. 2.

Soil loss rate estimate

The RUSLE (Renard et al. 1997) is the method most widely used worldwide to predict soil loss from inter-rill (sheet) and rill erosion caused by rainfall and associated overland flow. The soil loss rate is obtained as the product of two categories of factors that, respectively, represent the

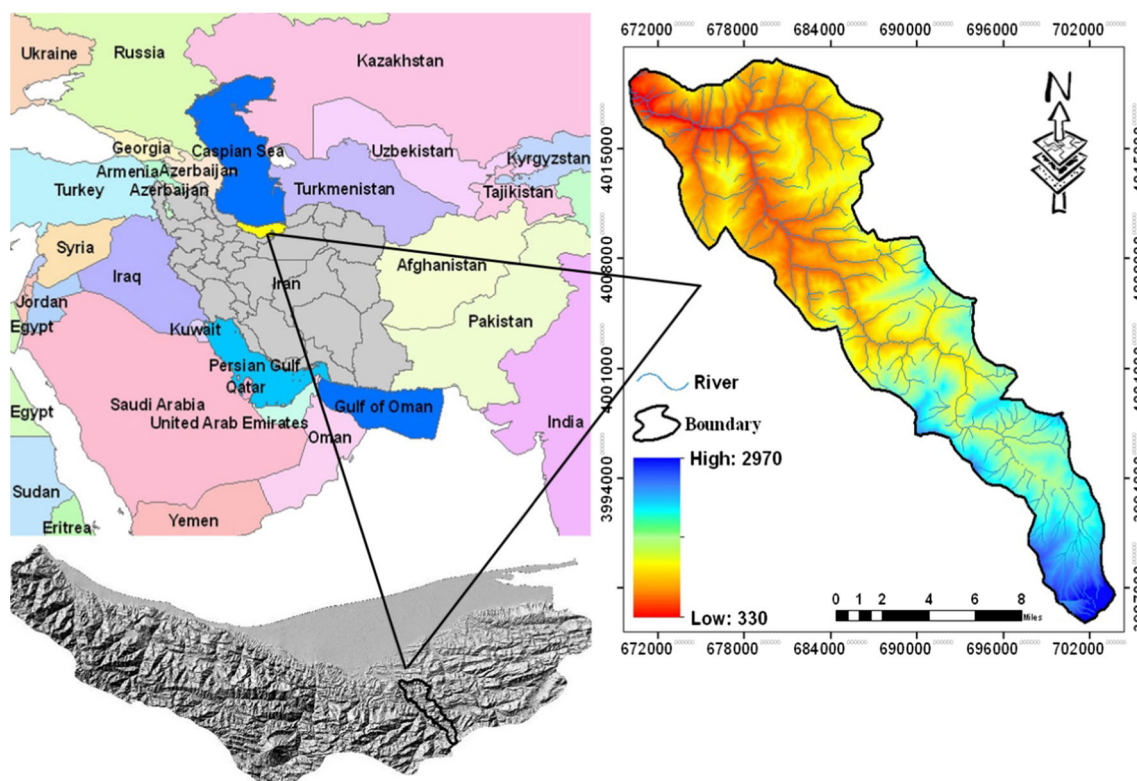


Fig. 1 Study area map

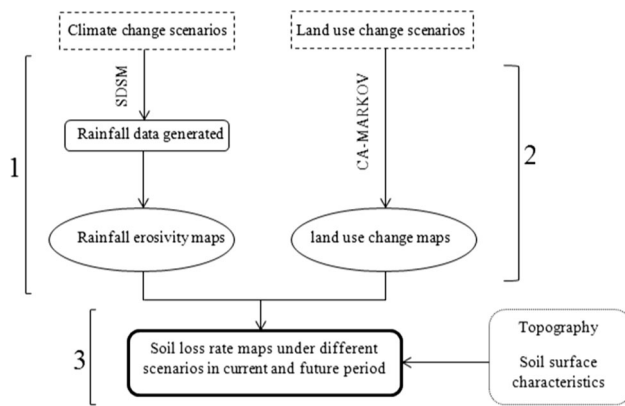


Fig. 2 Methodological flow chart of the study

natural and the human forces that act in soil erosion processes (Eq. 1):

$$A = (R \times K \times LS) \times (C \times P) \quad (1)$$

where A is the average soil loss caused by erosion ($t \text{ ha}^{-1} \text{ year}^{-1}$), K is the soil erodibility factor ($t \text{ ha h MJ}^{-1} \text{ ha}^{-1} \text{ mm}^{-1}$), L is the slope length factor, and S is the slope steepness; the first category that accounts for the R is the rainfall erosivity factor ($\text{MJ mm ha}^{-1} \text{ h}^{-1} \text{ year}^{-1}$). The other category of factors accounts for the effects of human activity on soil stability and conservation, expressed via C , which is the cover and management practice factor; P is the conservation support practice factor. The L , S , C , and P factors are dimensionless.

Generally speaking, RUSLE has some limitations; first of all, soil erosion by RUSLE model takes into consideration of various factors. The effect level of these factors may change the type and dimension of erosion. Moreover, the second problem of RUSLE model is that it predicts only soil erosion and RUSLE does not predict sediment deposition. Also based on results of Stanchi et al. (2014) this model has some error to simulate soil erosion in the winter time due to snow cover.

Rainfall erosivity factor (R): R is an index of the potential erosivity, which is determined as a function of the frequency, intensity, and duration of rainfall (Xu et al. 2012). In the present study, daily rainfall data of 20 years (1991–2010) were used to calculate the rainfall erosivity index and R defined based on maximum rainfall intensity; this factor is calculated with two parameters of total rainfall kinetic energy and 30-min maximum rainfall (Eq. 2):

$$R = \frac{1}{n} \sum_{i=1}^n \left[\sum_{k=1}^m \text{KE}(I_{30}) \right] \quad (2)$$

where R is rainfall erosivity ($\text{MJ mm ha}^{-1} \text{ h}^{-1} \text{ year}^{-1}$), KE is the total kinetic energy of each shower (MJ ha^{-1}), and I_{30} is the maximum intensity of 30-min rainfall (mm h^{-1})

(Renard et al. 1997). In addition, the kinetic energy of each shower is calculated via the Wischmeier and Smith (1978) (Eq. 3):

$$\text{KE} = (11.98 + 8.73 \log_{10} I) \quad (3)$$

Soil erodibility factor (K): K factor indicates the resistance of soil to rainfall erosivity effect (Karamesouti et al. 2016). In this study, K factor incorporates the restrictions of the K nomograph to the K factor equation, thus leading to the following extended K factor equation that requires four steps of calculation: (Auerswald et al. 2014):

$$\begin{aligned} 1) \quad K_1 &= 2.77 \times 10^{-5} \times (f_{\text{si+vfsa}} \times (100 - f_{\text{cl}}))^{1.14} \\ &\quad \text{for } f_{\text{si+vfsa}} < 70\% \quad K_1 = 1.77 \times 10^{-5} \\ &\quad \times (f_{\text{si+vfsa}} \times (100 - f_{\text{cl}}))^{1.14} + 0.0024 \\ &\quad \times f_{\text{si+vfsa}} + 0.16 \quad \text{for } f_{\text{si+vfsa}} > 70\% \\ 2) \quad K_2 &= (12 - f_{\text{OM}})/10 \quad \text{for } f_{\text{OM}} < 4\% \\ K_2 &= 0.8 \quad \text{for } f_{\text{OM}} > 4\% \\ 3) \quad K_3 &= K_1 \times K_2 + 0.043 \times (A - 2) + 0.033 \times (P - 3) \\ &\quad \text{for } K_1 \times K_2 > 0.2 \\ K_3 &= 0.091 - 0.34 \times K_1 \times K_2 + 1.79 \times (K_1 \times K_2)^2 \\ &\quad + 0.24 \times K_1 \times K_2 \times A + 0.033 \\ &\quad \times (P - 3) \quad \text{for } K_1 \times K_2 < 0.2 \\ 4) \quad K &= K_3 \quad \text{for } f_{\text{rf}} < 1.5\% \\ K &= K_3 \times (1.1 \times \exp(-0.024 \times f_{\text{rf}}) - 0.06) \\ &\quad \text{for } f_{\text{rf}} > 1.5\% \end{aligned} \quad (4)$$

where $f_{\text{si+vfsa}}$: mass fraction (in %) of silt plus very fine sand $\text{Si} + \text{vfSa}$ ($2 \dots 100 \mu\text{m}$) in the fine earth fraction, f_{cl} : mass fraction (in %) of clay ($< 2 \mu\text{m}$) in the fine earth fraction, f_{OM} : mass fraction (in %) of organic matter in the fine earth fraction, A : soil structure index ($1 \dots 4$) increasing from very fine granular to blocky, platy, or massive (classes are defined in Wischmeier 1971), and C : permeability index ($1 \dots 6$) increasing from rapid to very slow (classes are defined in Wischmeier 1971).

Topographic factor (LS): LS factor is calculated by multiplying the L (length factor) and S (steepness factor) factors together (Moore and Burch 1986), which indicates the effect of topography on erosion (Yildirim 2012). In this study, the equation of Moore and Burch (1986), which is adopted for estimates of erosion at the catchment scale, was used.

$$L = 1.4[A_s/22.13]^{0.4} \quad (5)$$

$$S = [\sin \beta / 0.0896]^{1.3} \quad (6)$$

where A_s is the area of plot per unit width and β is the slope angle, computed from the DEM.

Cover management factor (*C*): *C* factor is defined as the land cover and management factor and indicates the importance of cropping and management practices on soil loss rates (Xu et al. 2012). Land use classification was done using freely downloadable Landsat images. The normalized difference vegetation index (NDVI), an index of the vegetation density and land cover health, is used along with the following formula (Eq. 7). To generate the *C* factor value image for the region (Prasannakumar et al. 2013).

$$C = \exp \left[-\alpha \frac{\text{NDVI}}{(\beta - \text{NDVI})} \right] \quad (7)$$

where α and β are unit less parameters that determine the shape of the curve that relates to NDVI and the *C* factor. Van der Knijff et al. (2000) found that this scaling approach provided better results than did assuming a linear relationship; the values of 2 and 1 were selected for the parameters α and β , respectively.

The conservation practice factor (*P*): *P* indicates the ratio between the soil losses expected for a certain soil conservation practice and that of up and down the slope plowing; this is used to value the positive effects of those support practices (Renard et al. 1997). As such, there are no conservation practices in the study area; however, slope can be used to estimate the *P* factor. *P* factor values increase where slope length and steepness are higher. Given that the same crop or land use practice the *P* factor depends on the slope, we used a variable *P* factor based on slope (Table 1). In fact, lower slope will reduce the overall conservation practices.

Land use prediction and scenarios

The CA–Markov equation was used to simulate changes in the future spatial–temporal pattern of land use in the Kasilian watershed. CA–Markov model is a hybrid model from Markov chain and cellular automata (Sang et al. 2011). Both the models are among the dynamic models in which space and time are considered discrete (Arsanjani et al. 2013). The main problem of Markov model is the lack of spatial dimension in the prediction of changes; thus, when this model is combined with the cellular automata, the capability of positioning spatial change patterns will be included in the model. In this study, a cellular automata–Markov hybrid model was used in IDRISI Kilimanjaro,

which contains both Markov and cellular automata models (Alsharif and Pradhan 2014, 2015). IDRISI software illustrates a technique of CA–Markov since a Markov chain analysis is performed in order to estimate the transition matrix between the two past dates and to estimate probabilities of change for the third date to be predicted. Then, CA estimates the spatial distribution of land use at a later date. Equations (8) and (9) below illustrate the evaluation of cells from time *t* to *t* + 1 is determined by a function of its state; its neighborhood space; and a set of transition rule (Samat et al. 2011).

$$LU_{ij(t+1)} = f(LU_{ij(t)} \cdot S_{ij(t)} \cdot P_{x,y,ij(t)} \cdot N_{ij(t)}) \quad (8)$$

$$N_{ij(t)} = \frac{\sum_t N_{ij(t)}}{\text{\#of adjacent cells}} \quad (9)$$

where $LU_{ij(t+1)}$: the potential of cell *i, j* to change at time *t* + 1, $LU_{ij(t)}$: states of cell *i, j* at time *t*, $S_{ij(t)}$: suitability indexes of cell *i, j* at time *t*, $P_{x,y,ij(t)}$: probability of cell *i, j* to change from state *x* to state *y* at time *t*, $N_{ij(t)}$: neighborhood index of cell *i, j*.

The MCE method was used to produce suitability maps. These maps were extracted from many factors based on soil conservation, such as rainfall, topography factors, population density, distance from road, settlements, as well as their influence on land use. A topographic map with 1:25,000 scale was used to create a DEM with the 30*30 m pixel size. Slope, aspect and elevation maps were created from DEM using ArcGIS 9.3 software. Also drainage network and road maps were extracted from topographic maps, and distance from these phenomena was calculated in GIS. Digital map of soil textures and geology was prepared from the natural resources head office located at Mazandaran Province. Those factors were identified and quantified into one suitability map based on certain weights; these were standardized from 0 to 255. In fact, higher values in CA transition indicate greater suitability (Poska et al. 2008). One of the advantages of suitability map using the MCE method is determining constrains and factors in each land use development in future that was used as the first scenario in this study. In the second scenario, suitability maps were automatically prepared by IDRISI software, based on the trend of land use demand.

The validation model was used to assess the level of agreement between the observed 2011 and simulated 2011 land use datasets based on the Kappa spatial correlation statistic (Mitsova et al. 2011) (Eq. 10).

$$K = \frac{P_o - P_c}{1 - P_c} \quad (10)$$

P_c and P_o are chance agreement and overall accuracy, respectively (Congalton and Green 2009). The kappa

Table 1 Attribute value of *P* factor

Slope (%)	<i>P</i> factor
0–2	0.5
2–12	0.6
12–16	0.7
16–20	0.8

coefficient value was well above 0.80, thereby demonstrating that the two datasets agreed very well (Zhang et al. 2011). Therefore, the transition probability matrix can be used to simulate the distribution pattern of land use cover datasets in the study area. As regards change detection maps between land use maps, the NDVI value in change areas was determined. Thus, between each type of land use conversion, a regression relationship for each pixel was established. Based on the type of land use change in future, land use simulation map was prepared by the CA–Markov method for 2030. Then, the location of land use change areas was identified from 2011 to 2030 and these areas were extracted from change detection map. In the next step, change areas were overlaid on the NDVI image for 2011. Depending on the type of land use change, the regression relationship was then applied. In fact, these areas show the extent of change for the NDVI index in 2030. In addition, the other areas that did not differ in the change detection map have constant and unchanging NDVI.

Future climate change

Statistical downscaling model (SDSM), which was developed by Wilby et al. (2002), is a hybrid of multiple linear regression that establishes a statistical/empirical relationship between National Centers for Environmental Prediction (NCEP), large-scale variables, and local-scale variables, and produces certain regression parameters. The SDSM modeling of daily precipitation consists of stepwise process: modeling precipitation occurrence followed by modeling of precipitation amounts while precipitation occurs (Dharmarathna et al. 2014):

$$O_i = \alpha_0 + \sum_{j=1}^n \alpha_j p_{ij} \quad (11)$$

$$R_i^{0.25} = \beta_0 + \sum_{j=1}^n \beta_j p_{ij} + e_i \quad (12)$$

where O_i is the daily precipitation occurrence, R_i is daily precipitation amount, p_{ij} are predictors, n is number of predictor, α and β are model factors, and e_i is modeling error.

Application of the SDSM requires two types of daily data that included observed data from local predictor and large-scale atmospheric predictor data [e.g., NCEP and global climate mode (GCM) data] from the grid box of the predictor dataset closest to the study area (Hashmi et al. 2011). The SDSM is structured by monthly model for both daily temperature and precipitation downscaling, in which 12 regression equations were derived using different regression parameters for each month equation that corresponded to 12 months. The output of SDSM is daily series,

when the model is established; NCEP and GCM based on daily data are used to construct the current and future daily weather series. This series mainly contains five parts: (1) selection of predicts and predictors, (2) model calibration, (3) weather generator, (4) generation of future series of climate variables, and (5) the scenario of rainfall trend based on three scenarios of 10% increase in rainfall, 10% decrease in rainfall, and unchanged rainfall.

A simulation of mean monthly rainfall during the calibration and validation of the SDSM was checked using the coefficient of correlation (R) and root-mean-square error (RMSE), difference (BIAS) and is defined as:

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum (S_i - O_i)^2} \quad (13)$$

$$\text{BIAS} = \frac{\sum_{i=1}^n (S_i - O_i)}{n} \quad (14)$$

$$R^2 = \left[\frac{\sum_{i=1}^n (S_i - \bar{S})(O_i - \bar{O})}{\sigma_S - \sigma_O} \right]^2 \quad (15)$$

where S_i and O_i are the values observed and estimated by the model for each station and n is the total number of evaluated samples. In addition, σ is the variance and $\bar{O}$ and $\bar{S}$ are the observation and simulation averages, respectively (Hassan et al. 2014).

Results and discussion

RUSLE factors in current period

The rainfall erosivity factor shows that the value of R factor for the years 1991–2010 ranges from 179 to 272 MJ mm ha⁻¹ h⁻¹ year⁻¹ (Fig. 3). The mean value of annual R factor is observed to be 205 MJ mm ha⁻¹ h⁻¹ year⁻¹, which implies that the rainfall in the study area was unstable and obviously fluctuated. The figure of R factor exhibits obvious spatial difference (Fig. 3). One can observe that R values decreased from northwest to southeast depending on precipitation characteristics. LS values are directly related to surface relief. The high values were along the river courses and in the mountainous areas (Fig. 3). The average value for this area was 31.6, and the highest LS value for the Kasilian watershed was calculated as 55.23; such a value indicates that the LS values were relatively unevenly distributed. K values ranged from 0.32 to 0.44 t ha h MJ⁻¹ ha⁻¹ mm⁻¹; the map of K values was generated to show spatial distribution of erodibility (Fig. 3). Based on the 2011 NDVI map, C value mean was 0.43 in the study area that indicated the extent of forest area compared to other land uses. Moreover, the greatest C value belongs to rangeland areas with poor conditions and the surroundings of settlement areas.

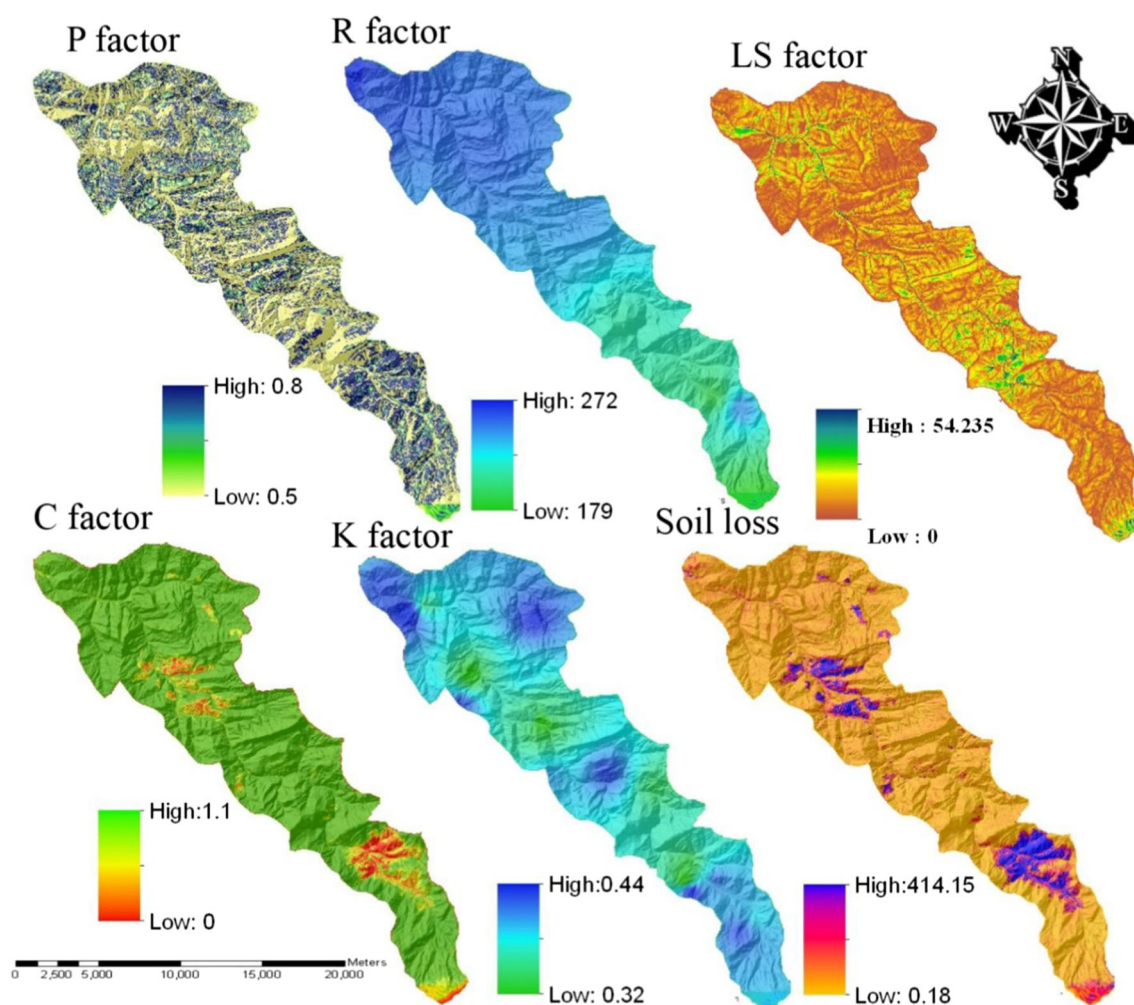


Fig. 3 Soil loss rate and factors of RUSLE model

Calibration and validation of SDSM

SDSM was first calibrated using large-scale predictor variables of the current climate condition derived from the NCEP re-analysis dataset as driving data and then validated in an independent period using NCEP data (Chu et al. 2010). In this study, the data on precipitation observed were used to calibrate the model parameters from 1961 to 1990, and the data for the period 1991–2000 were used to validate the model parameters by comparing the model output with observational data. Table 2 indicates that the accuracy evaluation was assessed based on R^2 , RMSE, and BIAS indices and for a 30-year period in six stations. Results show

an agreement between the simulated and observed amounts and low amounts of RMSE and BIAS. High amount of R^2 in different stations shows the same agreement and indicates that SDSM has reasonable power in downscaling the NCEP data. To validate the SDSM, the comparisons of monthly mean of the simulated and observed rainfall in different stations are shown in Fig. 4. Good matches occur between monthly mean of the simulated and observed precipitation. Therefore, the mentioned model can be used to generate the climatic data over the coming period. This study has shown SDSM is a tool that is able to generate daily weather data under different scenarios, based on expert opinions in present and future periods.

Table 2 Modeled monthly precipitation of SDSM in validation period

Statistical factor	Darzikola	Kaleh	Rig Cheshmeh	Sangdeh	Shirgah	Talar
BIAS	5.06	11.22	9.3	10.89	4.83	8.73
RSME	3.86	6.55	5.1	6.97	3.54	5.51
R^2	0.643	0.395	0.45	0.39	0.74	0.54

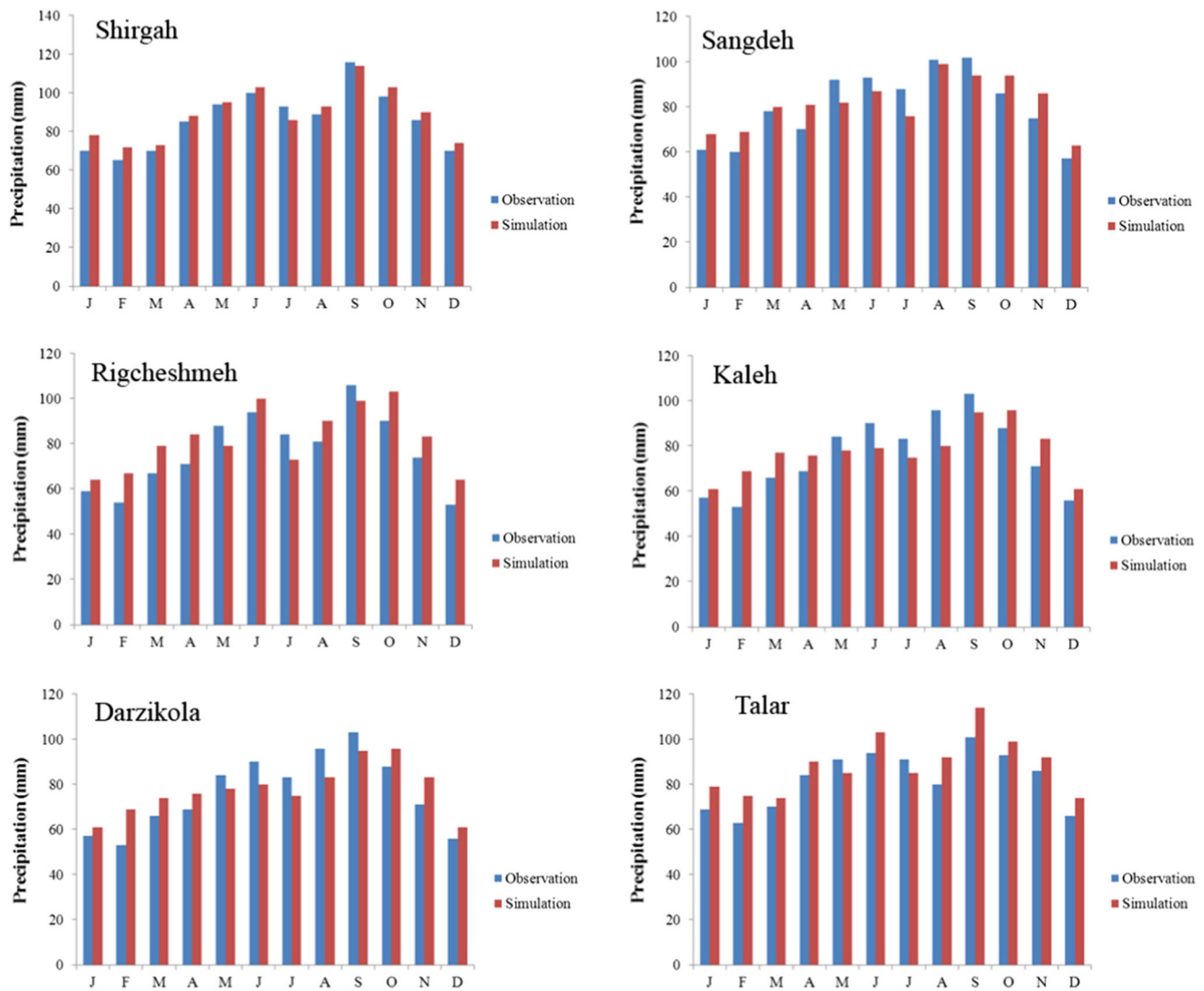


Fig. 4 Comparison of monthly mean precipitation among observed and downscaled derived from NCEP data during validation period

Impact of climate change scenarios on rainfall erosivity

Changes in precipitation

As regards the good performance of the SDSM-DC model in generating climate data using NCEP data, three scenarios of climate change have been derived for 2011–2030. The downscaling of the NCEP scenario to the scale of the Kasilian simulated a rainfall increase in 6–15% in the first and second scenarios and decrease in less than 1% in third

scenarios compared to the 1991–2010 period (Table 3). Based on NCEP downscale scenarios, rainfall will seemingly increase in the future period.

Changes in rainfall erosivity

The maximum and minimum of rainfall erosivity are for the first scenario of Talar station ($444.97 \text{ MJ mm ha}^{-1} \text{ h}^{-1} \text{ year}^{-1}$) and the second scenario of Darzikola station ($227.52 \text{ MJ mm ha}^{-1} \text{ h}^{-1} \text{ year}^{-1}$), respectively (Table 4). Furthermore, the results indicate that rainfall

Table 3 Current annual rainfall (mm) in the Kasilian watershed compared to SDSM-DC scenarios

Current rainfall	Future rainfall under SDSM-DC scenarios		
	10% increase in rainfall	10% decrease in rainfall	Unchanged in rainfall
822	945	765	867
Changes %	+15	−<1	6

Table 4 The amount of rainfall erosivity changes in different scenarios

Stations	Current period	Future period		
		Scenario 1	Scenario 2	Scenario 3
Darzikola	228.42	273	227.52	252.54
Kaleh	255.18	277.8	231.81	260.78
Rig Cheshmeh	277.62	306.03	255.07	283.38
Sangdeh	309.75	334.86	278.12	309.33
Shirgah	322.63	372.24	307.81	342.64
Talar	341.18	444.97	368.67	410.29
Mean	289.13	334.81	278.16	309.82

erosivity will increase in future compared with the current period so that if 10% decrease in rainfall scenario accords, rainfall erosivity will be slightly different from the current period, whereas the amount of rainfall will likely increase in future. In fact, the rainfall with extreme intensity and less duration has occurred and thus, rainfall erosivity will increase. Zhang et al. (2009) in China have estimated 20% increase in rainfall erosivity in three periods of 2020, 2050, and 2080. In addition, Segura et al. (2014) in the USA and Plangoen and Babel (2014) in Thailand have predicted 11 and 14% increase in rainfall erosivity in future. Results of Segura et al. (2014) explained that trends of erosivity vary widely spatially. Catchments in the northeastern and northwestern USA are characterized by strong increasing trends in R , while the trends in the midwestern and southwestern USA are either weak or inconsistent among the nine climatic projections considered. Also Plangoen and Babel (2014) showed that increase of erosivity will lead to 5.2 and 14.2 increase of erosion in the 2011–2040 and 2071–2099 periods.

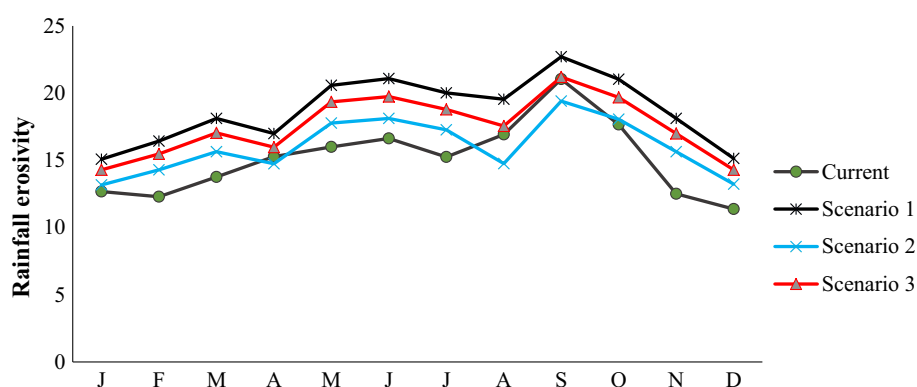
To investigate the role of different months of the year in the amount of rainfall erosivity and its effect on the erosion, different months in simulated periods and the current period were compared (Fig. 5). The total simulated rainfall erosivity in different months levels for the current and future periods is $11.38 \text{ MJ mm ha}^{-1} \text{ h}^{-1} \text{ year}^{-1}$ and $22.71 \text{ MJ mm ha}^{-1} \text{ h}^{-1} \text{ year}^{-1}$, respectively. Furthermore,

the results showed September and October are allocated the most erosivity and December has the least erosivity among all months in the year.

Future LULC change

The LULC data in 1986 and 2000 were operated by the Markov chain model to identify the probability of changing and transition areas. Such probability transition values were applied to predict the LULC for the year 2011 with the CA–Markov model. The result was the LULC map for 2011, as shown in Fig. 6. Based on the required preparation for the CA–Markov model, calibration map of year 2011 was plotted, as shown in Fig. 6. Standard Kappa value for the 2011 calibration map was equal to 0.93. Therefore, based on the acceptable level of validity of 2011, a calibration map was compared to the actual 2011 map, its suitability map, and land transition matrix based on factors and constraints in MCE model (Scenario 1). Furthermore, the trend of land use change as regards change detection (Scenario 2) was used to simulate the land use map of 2030 (Fig. 6). Results showed that forest use had the highest level of surface coverage in the classification of maximum likelihood method (Table 5). However, forest use has always been exposed to threats in the past 25 years such that its surface has decreased in this period, whereas the surface areas of residential and grassland use have increased. The detection matrix of the changes related to

Fig. 5 Comparison of monthly rainfall erosivity among current and downscaled derived from NCEP data in different scenarios



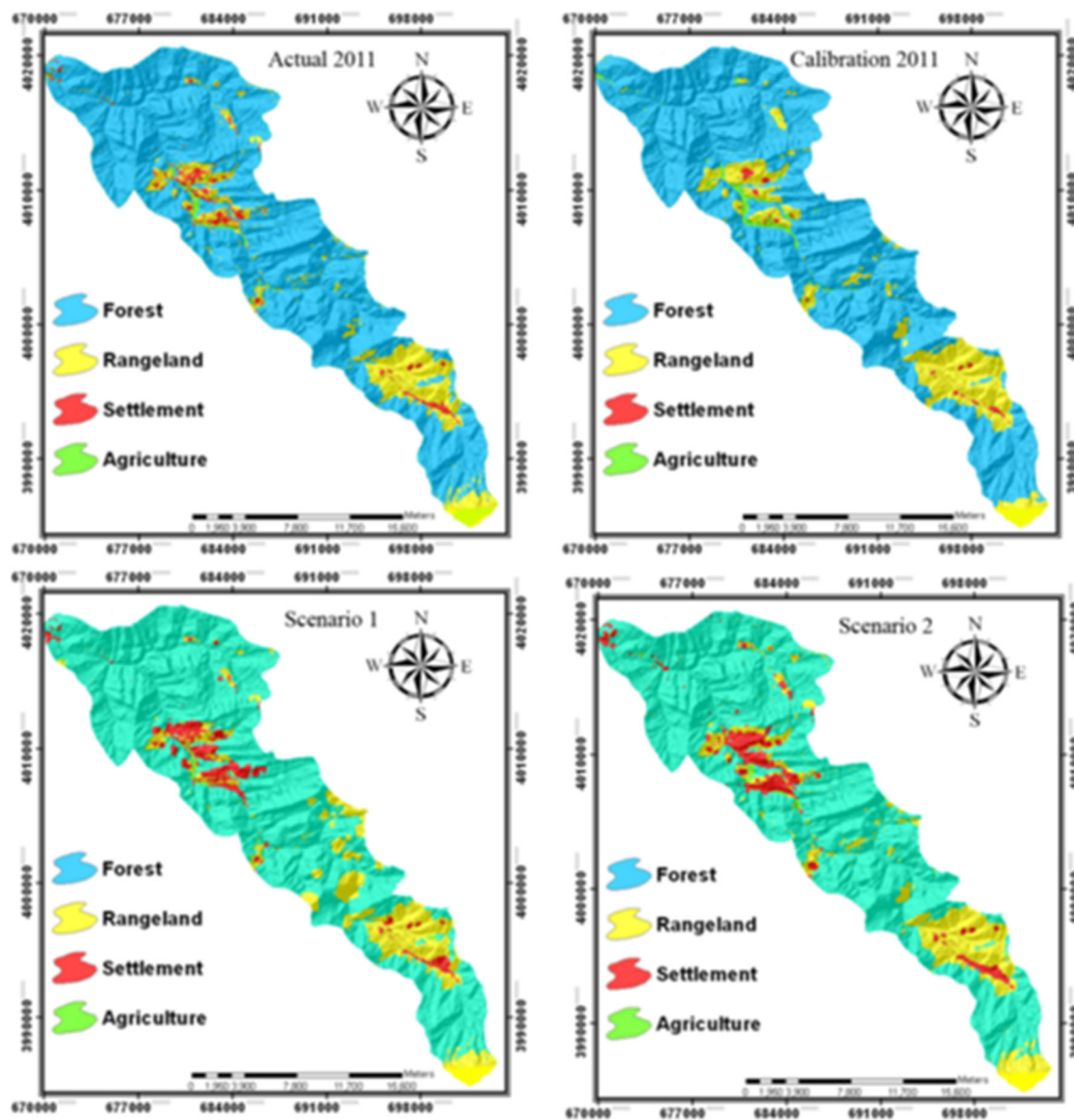


Fig. 6 Calibrated and actual land use/land cover maps for the year 2011 and simulated LULC maps for the years 2030

Table 5 Matrix for changes from year 2000 to year 2011

Land use	Forest	Rangeland	Settlement	Agriculture	Sum area
Forest	316,887	3210	0	337	320,434
Rangeland	8598	41,477	230	1138	51,443
Settlement	1127	230	2046	1675	5078
Agriculture	722	1138	4	2141	4005
Sum area	327,334	46,055	2280	5291	371,396

2000 to 2011 indicates that the highest levels of change and land use conversion were related to land use change from forest to grassland in the areas that surround forests. This pattern would continue in the future based on the map obtained for 2030. Results showed that because of the specific topographic condition in the region, land use

change from one class to the other was extremely hard. For instance, no considerable change occurred in the surface area of the agricultural land use during the past 25 years; neither would occur was well in the next 19 years. Specific properties of this basin, including sharp slopes in most parts of the region and low productivity of farm lands after

Table 6 The list of scenarios for simulating future soil loss rate under climate and land use change

Cod	Climate change scenarios	Land use change scenarios
<i>First</i>		
1	10% increase in rainfall for 2011–2030	Combination of CA–Markov and MCE
2	10% decrease in rainfall for 2011–2030	Combination of CA–Markov and MCE
3	Unchanged rainfall for 2011–2030	Combination of CA–Markov and MCE
4	10% increase in rainfall for 2011–2030	CA–Markov with automata suitability map
5	10% decrease in rainfall for 2011–2030	CA–Markov with automata suitability map
6	Unchanged rainfall for 2011–2030	CA–Markov with automata suitability map
<i>Second</i>		
7	10% increase in rainfall for 2011–2030	–
8	10% decrease in rainfall for 2011–2030	–
9	Unchanged rainfall for 2011–2030	–

a couple of agricultural activities, would not experience noticeable changes in its expansion or nature. Factors used in the construction of land suitability map showed that road plays a significant role in the land use change pattern such that residential sites in the north of the basin experienced more expansion because of road network that are more appropriate compared to the residential areas in the south of the basin. Therefore, the greater the expansion of road networks, the greater would be the degree of land use change.

Future soil loss rate under climate and land use change

In the RUSLE model, *K* factor, *LS*, and *P* factor will have little change in future. The aim of this research is investigating effects of land use and climate change on future soil loss; therefore, we did not change other input data of the model. The other factors of model were considered constant in the total period of research, so soil loss change is related to land use and climate changes. Thus, by simulating rainfall erosivity and *C* factor under effect of climate and land use change soil loss in future can be achieved. As regards several possibilities that may occur in climate condition and future land use, in this study, different land use and climate change scenarios based on CA–Markov and SDSM-DC were integrated with each other to be able to provide a suitable estimation of future erosion condition at the catchment scale. In this study, soil loss scenarios were divided into two groups. The first group scenarios are related to the simultaneous impact of land use and climate change on soil loss rate, comprising six scenarios. The second group scenarios concern the impact of land cover on erosion control that includes three scenarios. Therefore, *C* factor was removed from the RUSLE model and only climate change scenarios were considered, which include three scenarios. In fact, the first group scenarios are for soil loss intensity and the second group scenarios are for

potential soil loss. Thus, nine scenarios were developed in two groups as shown in Table 6; soil loss zonation was prepared by multiplying all effective factors in erosion (Fig. 7).

All of the soil loss scenario maps show a similar pattern, and the maps are similar in their spatial distribution of erosion rates, with a sharp contrast between the upstream and downstream areas (Fig. 7). Thus, mostly low erosion rates occur in the upstream area, whereas the downstream area has high erosion rates. In other words, the highest rate of soil loss is related to areas with the least land cover and the rangeland of villages surrounding. Extreme land degradation is caused by the development of settlement areas and abandoned agricultural lands because lack of soil fertility is a feature of these areas. Therefore, the soil without land vegetation and disrupted does not have any resistance against rainfall erosivity, which has led to extreme erosion in these areas. Despite the same pattern among all scenarios, soil loss rate has changed between current period and future scenarios. Thus, soil loss mean is $6 \text{ t ha}^{-1} \text{ year}^{-1}$ in current period and this period has the least amount of soil loss among all scenarios (Fig. 8). Furthermore, the type of scenarios has a direct impact on soil loss, so that if 10% increase in rainfall occurs, soil loss rate will not change much in both land use scenarios and it will increase 58% compared to current period. The results also show if climate change trend based on NCEP data and unchanged rainfall scenario are simultaneous, soil loss rate will be 47% more than the current period. A reason is that the rate was overcompensated by a distribution change that included higher rainfall erosivity in future. Several studies have emphasized the impact of rainfall erosivity on raising erosion when rainfall has not changed or has decreased (Neal et al. 2005; Shrestha et al. 2013; Plangoen et al. 2013; Simonneaux et al. 2015; Paroissein et al. 2015). Rainfall erosivity could induce ambivalent effects related to the interaction of land cover, plant biomass, erosion, runoff, and erosion (Nearing et al. 2005; Turnbull et al.

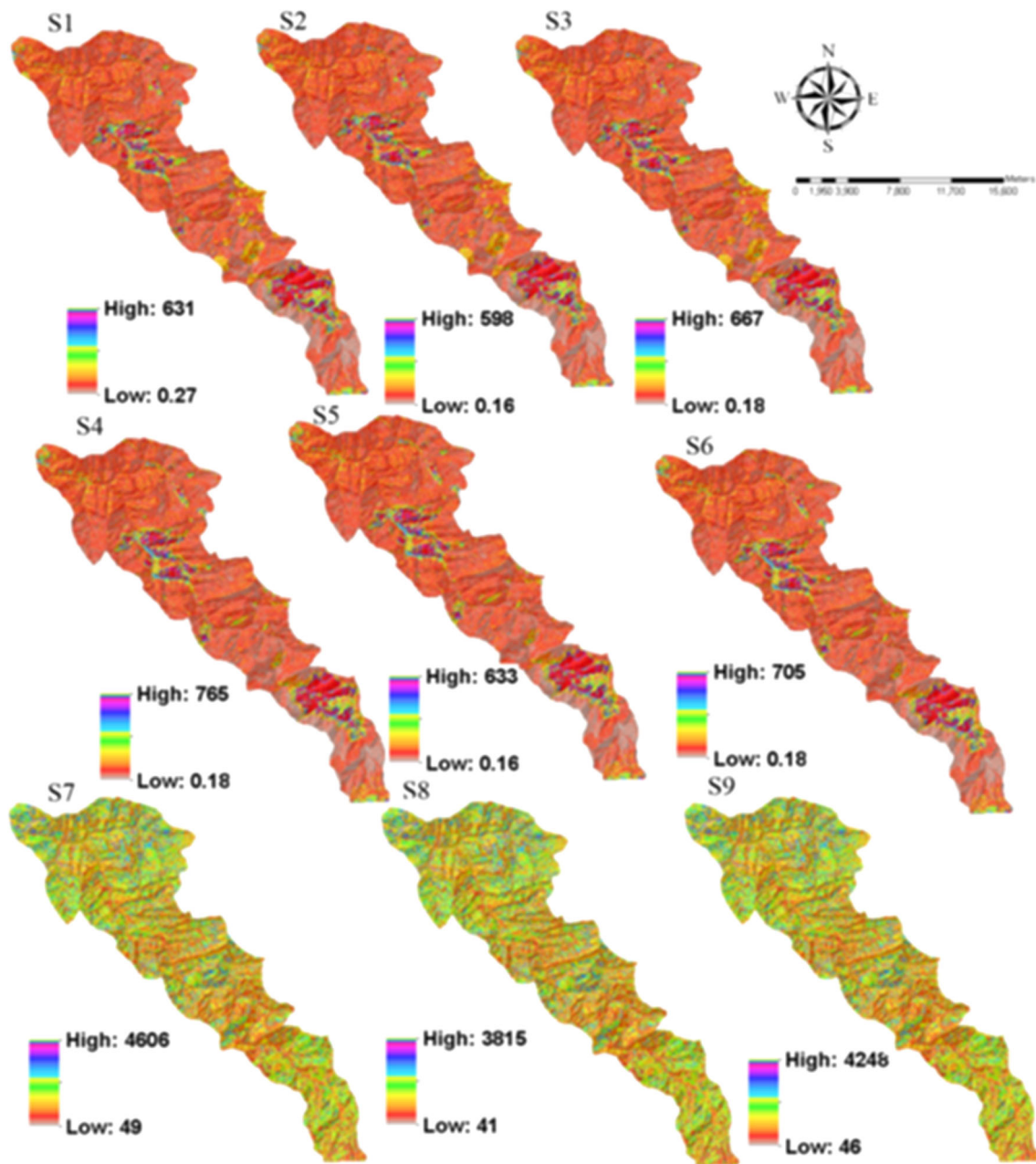


Fig. 7 Soil loss zonation maps under different scenarios in future period

2013). The results of soil loss scenarios indicate that the type of land use change scenarios has not made much difference in soil loss changes. Moreover, suitability maps in automata mode and multi-criteria do not have significant difference in soil loss. However, vegetation cover has an essential role in erosion control. A reason is that by removing C factor in RUSLE model, soil loss rate will greatly increase so that if soil loss mean is $9.17 \text{ t ha}^{-1} \text{ year}^{-1}$ in the first six scenarios, soil loss rate will be $88.48 \text{ t ha}^{-1} \text{ year}^{-1}$, by removing C factor. This indicates the positive role of land cover in erosion control.

Occupying 83% of the forest areas in the Kasilian watershed has caused soil loss rate to be much less than other watersheds in Iran, and C factor mean is 0.43. However, if land use change continues based on the scenarios, C value will increase 0.58 in 2030. The variation within the land use changes was approximately 23% for changes related only to forest or rangeland areas. These changes have caused a 0.15 increase in the amount of C , and therefore, the current sheet and inter-rill erosion will convert to rill and gully erosion and finally lead to rise in erosion. In addition to extensive changes in land use, increase in

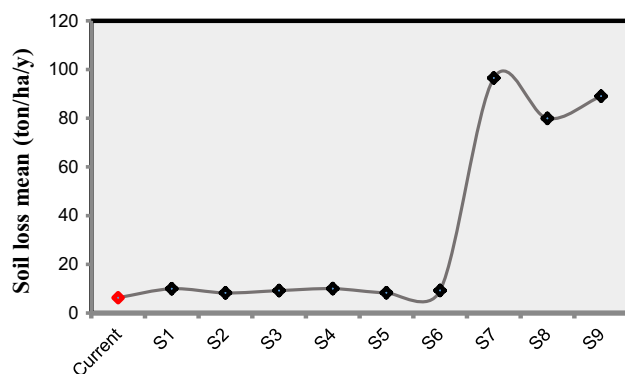


Fig. 8 Soil loss mean in current and future period under different scenarios

rainfall erosivity will have a great influence on intensifying soil erosion. Nearing et al. (2005) have emphasized that, based on the RUSLE model, a 1% increase in rainfall erosivity led to 1% increase in erosion. If unchanged rainfall scenario is continuous until 2030, the amount of rainfall erosivity will be 7% more than that of current period. Therefore, we expect that the soil loss rate will increase 7%. Soil loss rate under the unchanged rainfall scenario and future land use change was 47% higher than that of the current period, thereby indicating the strong role of land use change impact compared to rainfall erosivity change on soil loss rate. Plangoen et al. (2013) and Paroissien et al. (2015) have emphasized the more important role of land use than rainfall erosivity change on soil loss.

Conclusion

This study examined the role played by climate and land use changes on soil loss rate for the Kasilian watershed (north of Iran) over the period 1991–2030. Investigating soil loss rate until 2030 indicates that soil loss is thoroughly influenced by climate and land cover patterns in future. Climate change and vegetation change play an important role in soil loss by change within rainfall erosivity and land use, respectively. Rainfall erosivity in future climate under the output of downscaling NCEP data and under three different scenarios will be much more than that of the current period. These results show that rainfalls with less duration and more intensity will occur. The overall decrease in forest areas as well as increase in rangeland areas and fivefolded development of settlement areas will result in 35% rise of C value in the RUSLE model. Thus, simultaneity of alteration in rainfall patterns with land cover change will lead to a 31–58% increase in soil loss rate in future period compared to the current period. The results indicate that soil loss has been closely related to

land use change and that various scenarios of climate change have played a secondary role in the development of erosion. Climate change is a natural process and almost unavoidable. Thus, if land use change can be controlled, and with the existence of future climate change, soil loss can be greatly controlled and even reduced.

References

- Alkharabsheh M, Alexandridis TK, Bilas G, Misopolinos N, Silleos N (2013) Impact of land cover change on soil erosion hazard in northern Jordan using remote sensing and GIS. *Procedia Environ Sci* 19:912–921. doi:[10.1016/j.proenv.2013.06.101](https://doi.org/10.1016/j.proenv.2013.06.101)
- Alsharif AAA, Pradhan B (2014) Monitoring and predicting land use change in Tripoli Metropolitan City using an integrated Markov chain and cellular automata models in GIS. *Arab J Geosci* 7(10):4291–4301. doi:[10.1007/s12517-013-1119-7](https://doi.org/10.1007/s12517-013-1119-7)
- Alsharif AAA, Pradhan B (2015) Predicting the spatial patterns of urban expansion by combining the Chi squared automatic integration detection decision tree, markov chain, and cellular automata models in GIS. *Geocarto Int* 30(8):858–881. doi:[10.1080/10106049.2014.997308](https://doi.org/10.1080/10106049.2014.997308)
- Arsanjani J, Helbich M, Kainz W, Boloorani AD (2013) Integration of logistic regression, Markov chain and cellular automata models to simulate urban expansion. *Int J Appl Earth Obs Geoinf* 21:265–275
- Auerswald K, Fiener P, Martin W, Elhaus D (2014) Use and misuse of the K factor equation in soil erosion modeling: An alternative equation for determining USLE nomograph soil erodibility values. *CATENA* 118:220–225. doi:[10.1016/j.catena.2014.01.008](https://doi.org/10.1016/j.catena.2014.01.008)
- Biro K, Pradhan B, Buchroithner MF, Makeschin F (2013) An assessment of land use/land-cover change impacts on soil properties in the northern part of Gadarif region, Sudan. *Land Degrad Dev* 24(1):90–102. doi:[10.1002/ldr.1116](https://doi.org/10.1002/ldr.1116)
- Chen T, Niu R, Li P, Zhang L, Du B (2011) Regional soil erosion risk mapping using RUSLE, GIS, and remote sensing: a case study in Miyun Watershed, North China. *Environ Earth Sci* 63:533–541. doi:[10.1007/s12665-010-0715-z](https://doi.org/10.1007/s12665-010-0715-z)
- Chu JT, Xia J, Xu CY, Singh VP (2010) Statistical downscaling of daily mean temperature, pan evaporation and precipitation for climate change scenarios in Haihe River, China. *Theor Appl Climatol* 99:149–161. doi:[10.1007/s00704-009-0129-6](https://doi.org/10.1007/s00704-009-0129-6)
- Congalton RG, Green K (2009) Assessing the accuracy of remotely sensed data: principles and practice, 2nd edn. CRC/Taylor & Francis, Boca Raton, p 183p
- Dharmarathna WRSS, Herath S, Weerakoon SB (2014) Changing the planting date as a climate change adaptation strategy for rice production in Kurunegala district, Sri Lanka. *Sustain Sci* 9:103–111
- Dibike YB, Coulibaly P (2005) Hydrologic impact of climate change in the Saguenay watershed: comparison of downscaling methods and hydrologic models. *J Hydrol* 307(1–4):145–163
- Hashmi MZ, Shamseldin AY, Melville BW (2011) Comparison of SDSM and LARS-WG for simulation and downscaling of extreme precipitation events in a watershed. *Stoch Environ Res Risk Assess* 25(4):475–484. doi:[10.1007/s00477-010-0416-x](https://doi.org/10.1007/s00477-010-0416-x)
- Hassan Z, Shamsudin S, Harun S (2014) Application of SDSM and LARS-WG for simulating and downscaling of rainfall and temperature. *Theor Appl Climatol* 116:243–257. doi:[10.1007/s00704-013-0951-8](https://doi.org/10.1007/s00704-013-0951-8)

- Hoomehr S, Schwartz J, Yoder D (2016) Potential changes in rainfall erosivity under GCM climate change scenarios for the southern Appalachian region, USA. *CATENA* 136:141–151. doi:[10.1016/j.catena.2015.01.012](https://doi.org/10.1016/j.catena.2015.01.012)
- Imeson AC, Lavee H (1998) Soil erosion and climate change: the transect approach and the influence of scale. *Geomorphology* 23:219–227. doi:[10.1016/S0169-555X\(98\)00005-1](https://doi.org/10.1016/S0169-555X(98)00005-1)
- Jiang L, Yao Z, Liu Z, Wu S, Wang R, Wang L (2015) Estimation of soil erosion in some sections of Lower Jinsha River based on RUSLE. *Nat Hazards* 76:1831–1847. doi:[10.1007/s11069-014-1569-6](https://doi.org/10.1007/s11069-014-1569-6)
- Karamesouti M, Petropoulos G, Papanikolaou D, Kairis O, Kosmas K (2016) Erosion rate reductions from PESERA and RUSLE at a Mediterranean site before and after a wildfire: comparison and implications. *Geoderma* 261:44–58. doi:[10.1016/j.geoderma.2015.06.025](https://doi.org/10.1016/j.geoderma.2015.06.025)
- Khosrokhani M, Pradhan B (2014) Spatio-temporal assessment of soil erosion at Kuala Lumpur metropolitan city using remote sensing data and GIS. *Geomat Nat Haz Risk* 5(3):252–270. doi:[10.1080/19475705.2013.794164](https://doi.org/10.1080/19475705.2013.794164)
- Klike A, Eitzinger J (2010) Impact of climate change on soil erosion and the efficiency of soil conservation practices in Austria. *J Agric Sci* 148:529–541. doi:[10.1017/S0021859610000158](https://doi.org/10.1017/S0021859610000158)
- Li Q, Yu P, Li G, Zhou D, Chen X (2014) Overlooking soil erosion induces underestimation of the soil C loss in degraded land. *Quatern Int* 394:287–290. doi:[10.1016/j.quaint.2014.05.034](https://doi.org/10.1016/j.quaint.2014.05.034)
- Litschert S, Theobald D, Brown T (2014) Effects of climate change and wildfire on soil loss in the Southern. *CATENA* 118:206–219. doi:[10.1016/j.catena.2014.01.007](https://doi.org/10.1016/j.catena.2014.01.007)
- Liu L, Liu Z, Ren X, Fischer T, Xu Y (2011) Hydrological impacts of climate change in the Yellow River Basin for the 21st century using hydrological model and statistical downscaling model. *Quat Int* 244(2):211–220
- Maeda E, Petri K, Siljander M, Clark B (2010) Potential impacts of agricultural expansion and climate change on soil erosion in the Eastern Arc Mountains of Kenya. *Geomorphology* 123:279–289
- Mitsova D, Shuster W, Wang XH (2011) A cellular automata model of land cover change to integrate urban growth with open space conservation. *Landsc Urban Plan* 99:141–153. doi:[10.1016/j.landurbplan.2010.10.001](https://doi.org/10.1016/j.landurbplan.2010.10.001)
- Mondal A, Khare D, Kundu S, Meena P, Mishra P, Shukla R (2015) Impact of climate change on future soil erosion in different slope, land use, and soil-type conditions in a part of the Narmada River Basin, India. *J Hydrol Eng.* doi:[10.1061/\(ASCE\)HE.1943-5584.0001065](https://doi.org/10.1061/(ASCE)HE.1943-5584.0001065)
- Moore ID, Burch GJ (1986) Physical basis of the length-slope factor in the universal soil loss equation. *Soil Sci Soc Am J* 50:1294–1298. doi:[10.2136/sssaj1986.03615995005000050042x](https://doi.org/10.2136/sssaj1986.03615995005000050042x)
- Mullan D, Favis-Mortlock D, Fealy R (2012) Addressing key limitations associated with modeling soil erosion under the impacts of future climate change. *Agric For Meteorol* 156:18–30. doi:[10.1016/j.agrformet.2011.12.004](https://doi.org/10.1016/j.agrformet.2011.12.004)
- Neal M, Nearing MA, Vining R, Southworth J, Pfeifer A (2005) Climate change impacts on soil erosion in Midwest United States with changes in crop management. *CATENA* 61:165–184. doi:[10.1016/j.catena.2005.03.003](https://doi.org/10.1016/j.catena.2005.03.003)
- Nearing AM (2001) Potential changes in rainfall erosivity in the U.S. with climate change during the 21st century. *J Soil Water Conserv* 56(3):229–232
- Nearing MA, Jetten V, Baffaut C (2005) Modeling response of soil erosion and runoff to changes in precipitation and cover. *CATENA* 61(2–3):131–154. doi:[10.1016/j.catena.2005.03.007](https://doi.org/10.1016/j.catena.2005.03.007)
- Pan J, Wen Y (2014) Estimation of soil erosion using RUSLE in Caijiamiao watershed, China. *Nat Hazards* 71:2187–2205. doi:[10.1007/s11069-013-1006-2](https://doi.org/10.1007/s11069-013-1006-2)
- Paroissein J, Darboux F, Couturier A, Devillers B, Mouillot F, Raclot D, Bissonnais Y (2015) A method for modeling the effects of climate and land use changes on erosion and sustainability of soil in a Mediterranean watershed (Languedoc, France). *J Environ Manag* 150:57–68. doi:[10.1016/j.jenvman.2014.10.034](https://doi.org/10.1016/j.jenvman.2014.10.034)
- Plangoen P, Babel M (2014) Projected rainfall erosivity changes under future climate in the Upper Nan Watershed, Thailand. *J Earth Sci Clim Change* 5(10):1–7
- Plangoen P, Babel M, Clemente R, Shrestha S, Tripathi N (2013) Simulating the impact of future land use and climate change on soil erosion and deposition in the mae nam nan sub-catchment, Thailand. *Sustainability* 5:3244–3274. doi:[10.3390/su5083244](https://doi.org/10.3390/su5083244)
- Poska A, Sepp E, Veski S, Koppel K (2008) Using quantitative pollenbased land-cover estimations and a spatial CA Markov model to reconstruct the development of cultural landscape at Rouge, South Estonia. *Veg Hist Archaeobot* 17:527–541. doi:[10.1007/s00334-007-0124-8](https://doi.org/10.1007/s00334-007-0124-8)
- Pradhan B, Chaudhari A, Adinarayana J, Buchroithner MF (2012) Soil erosion assessment and its correlation with landslide events using remote sensing data and GIS: a case study at Penang Island, Malaysia. *Environ Monit Assess* 184(2):715–727. doi:[10.1007/s10661-011-1996-8](https://doi.org/10.1007/s10661-011-1996-8)
- Prasannakumar V, Vijith H, Abinod S, Geetha N (2013) Estimation of soil erosion risk within a small mountainous sub-watershed in Kerala, India, using Revised Universal Soil Loss Equation (RUSLE) and geo-information technology. *Geosci Front* 3(2):209–215. doi:[10.1016/j.gsf.2011.11.003](https://doi.org/10.1016/j.gsf.2011.11.003)
- Pruski FF, Nearing MA (2002) Climate-induced changes in erosion during the 21st century for eight U.S. locations. *Water Resour Res* 38(12):34–44
- Quan B, Romkens M, Li R, Wang F, Chen J (2011) Effect of land use and land cover change on soil erosion and the spatio-temporal variation in Liupan Mountain Region, southern Ningxia, China. *Front Environ Sci Eng* 5(4):564–572. doi:[10.1007/s11783-011-0348-9](https://doi.org/10.1007/s11783-011-0348-9)
- Ranzi R, Le T, Rulli M (2012) A RUSLE approach to model suspended sediment load in the Lo river (Vietnam): effects of reservoirs and land use changes. *J Hydrol* 422–423:17–29. doi:[10.1016/j.jhydrol.2011.12.009](https://doi.org/10.1016/j.jhydrol.2011.12.009)
- Renard KG, Foster GA, Weesies DA, McCool DK, Yoder DC (1997) Predicting soil erosion by water: a guide to conservation planning with the revised universal soil loss equation (RUSLE). *Agriculture Handbook*. USDA, Washington, DC, p 557
- Rizee HM, Saharkhiz MA, Pradhan B, Ahmad N (2016) Soil erosion prediction based on land cover dynamics at the Semenyih watershed in Malaysia using LTM and USLE models. *Geocarto International* 31(10):1158–1177. doi:[10.1080/10106049.2015.1120354](https://doi.org/10.1080/10106049.2015.1120354)
- Routschek A, Schmidt J, Kreienkamp F (2014) Impact of climate change on soil erosion—a high-resolution projection on catchment scale until 2100 in Saxony/Germany. *CATENA* 121:99–109. doi:[10.1016/j.catena.2014.04.019](https://doi.org/10.1016/j.catena.2014.04.019)
- Samat N, Hasni R, Elhadari YAE (2011) Modelling land use changes at the peri-urban areas using GEO-GRAPHIC information systems and cellular automata model. *J Sustain Dev* 4(6):72–84
- Sang L, Zhang C, Yang J, Zhu D, Yun W (2011) Simulation of land use spatial pattern of towns and villages based on CA–Markov model. *Math Comput Model* 54:938–943
- Segura C, Sun G, McNulty S, Zhang Y (2014) Potential impacts of climate change on soil erosion vulnerability across the conterminous United States. *J Soil Water Conserv* 69:171–181. doi:[10.2489/jswc.69.2.171](https://doi.org/10.2489/jswc.69.2.171)
- Sharma A, Tiwari K, Bhadoria PBS (2011) Effect of land use land cover change on soil erosion potential in an agricultural watershed. *Environ Monit Assess* 173:789–801. doi:[10.1007/s10661-010-1423-6](https://doi.org/10.1007/s10661-010-1423-6)

- Shrestha B, Babel MS, Maskey S, Griensven AV, Uhlenbrook S, Green A, Akkharath I (2013) Impact of climate change on sediment yield in the Mekong River basin: a case study of the Nam Ou basin, Lao PDR. *Hydrol Earth Syst Sci* 17:1–20. doi:[10.5194/hess-17-1-2013](https://doi.org/10.5194/hess-17-1-2013)
- Simonneaux V, Cheggour A, Deschamps C, Mouillot F, Cerdan O, Bissonnais Y (2015) Land use and climate change effects on soil erosion in a semi-arid mountainous watershed (High Atlas, Morocco). *J of Ari Environ* 122:64–75. doi:[10.1016/j.jaridenv.2015.06.002](https://doi.org/10.1016/j.jaridenv.2015.06.002)
- Stanchi S, Godone D, Freppaz M, Zanini E (2013) Assessing the susceptibility of alpine soils to erosion using soil physical and site indicators. *Soil Use Manag*. doi:[10.1111/sum.12063](https://doi.org/10.1111/sum.12063)
- Stanchi S, Freppaz M, Ceaglio E, Maggioni M, Meusburger K, Alewell C, Zanini E (2014) Soil erosion in an avalanche release site (Valle d'Aosta: Italy): towards a winter factor for RUSLE in the Alps. *Nat Hazards Earth Syst Sci* 14:1761–1771
- Turnbull L, Parsons AJ, Wainwright J, Anderson JP (2013) Runoff responses to long-term rainfall variability in a shrub-dominated catchment. *J Arid Environ* 91:88–94. doi:[10.1016/j.jaridenv.2012.12.002](https://doi.org/10.1016/j.jaridenv.2012.12.002)
- Van der Knijff JM, Jones RJA, Montanarella L (2000) Soil erosion risk assessment in Europe. EUR 19044 EN. Office for Official Publications of the European Communities. Luxembourg. 34 pp
- Wijitkosum S (2012) Impacts of land use changes on soil erosion in Pa Deng Sub-district, adjacent area of Kaeng Krachan National Park, Thailand. *Soil Water Resour* 7:10–17
- Wilby RL, Dawson CW, Barrow EM (2002) SDSM—a decision support tool for the assessment of regional climate change impacts. *Environ Model Softw* 17(2):145–157. doi:[10.1016/S1364-8152\(01\)00060-3](https://doi.org/10.1016/S1364-8152(01)00060-3)
- Wilson TS, Sleeter BM, Davis AW (2015) Potential future land use threats to California protected areas. *Reg Environ Ch* 15:1051–1064
- Wischmeier WH (1971) A soil erodibility nomograph for farmland and construction sites. *J Soil Water Conserv* 26:189–193
- Wischmeier W, Smith D (1978) Predicting rainfall erosion losses. A guide to conservation planning. USDA. Agr Res Serv Handbook. 537 pp
- Xu L, Xu X, Meng X (2012) Risk assessment of soil erosion in different rainfall scenarios by RUSLE model coupled with information diffusion model: a case study of Bohai Rim, China. *CATENA* 100:74–82. doi:[10.1016/j.catena.2012.08.012](https://doi.org/10.1016/j.catena.2012.08.012)
- Yildirim U (2012) Assessment of soil erosion at the Dedirmen Creek watershed area. Afyonkarahisar, Turkey, pp 73–80. doi: [10.5053/isepp.2011.1-7](https://doi.org/10.5053/isepp.2011.1-7)
- Zhang XC, Liu WZ, Zheng FL (2009) Simulating site-specific impacts of climate change on soil erosion and surface hydrology in southern Loess Plateau of China. *CATENA* 79:237–242. doi:[10.1016/j.catena.2009.01.006](https://doi.org/10.1016/j.catena.2009.01.006)
- Zhang RQ, Tang CJ, Ma SH, Yuan H, Gao LL, Fan WY (2011) Using Markov chains to analyze changes in wetland trends in arid Yinchuan Plain, China. *Math Comput Model* 54:924–930. doi:[10.1016/j.mcm.2010.11.017](https://doi.org/10.1016/j.mcm.2010.11.017)
- Zhang B, He C, Burnham M, Zhang L (2016) Evaluating the coupling effects of climate aridity and vegetation restoration on soil erosion over the Loess Plateau in China. *Sci Total Environ* 539:439–449. doi:[10.1016/j.scitotenv.2015.08.132](https://doi.org/10.1016/j.scitotenv.2015.08.132)

Reproduced with permission of copyright owner. Further reproduction
prohibited without permission.